

GRID-CLOCK QUANTUM GRAVITY

Finite Information, Emergent General Relativity, Regular Black Holes, and Unitary Evaporation

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SCOPE NOTE. This paper presents the gravity construction as one continuous candidate theory: finite microscopic information and capacity, finite-Weyl quantum kinematics, strict-local quantum-cellular-automaton propagation, exact microscopic update composition, emergent ADM/DeWitt gravity, finite-capacity strong gravity, regular black-hole cores, horizon boundary information, and unitary evaporation. Statements are explicitly separated into postulates, derived consequences, conditional completion assumptions, numerical or algebraic audits, semiclassical matching results, and genuinely open microscopic problems.

Abstract

Grid-Clock Quantum Gravity (GCQG) is a finite-information candidate theory in which a fixed relational graph and a universal microscopic update index are more fundamental than continuum spacetime. Each elementary capacity cell has bounded local energy, while finite information is represented separately by finite-dimensional Weyl records. Exactness is assigned to finite microscopic update composition rather than to the continuum infinitesimal hypersurface-deformation algebra at fixed cutoff; the ADM/DeWitt algebra is required to emerge in the smooth long-wavelength sector. Geometry-dependent normal deformation generates the inverse-metric H-H structure function, and numerical refinement audits support the H-D and D-D sectors. In the strong-gravity sector, the Planck-capacity condition yields a finite density bound and a core scale proportional to $M^{1/3}$; a two-channel Compute-Link closure and a finite-Weyl chord-load completion give a conditional Hayward-type regular profile. The present revision adds a strict one-neighborhood-per-update BCC quantum-cellular-automaton propagation candidate. In the minimal linear isotropic Weyl class, published quantum-walk classifications support the BCC/Weyl structure, while a doubled $d=4$ Clifford carrier supplies two positive-frequency propagation amplitudes. A local horizon normal then defines $P_{\text{out}}=(I+\alpha_n)/2$ with rank two for every orientation, so the effective outgoing boundary algebra is $M_2(\mathbb{C})$. Under a minimal boundary-record identification this selects $d_L=2$ rather than assuming a qubit by hand. Gauge-reduced boundary counting gives $2^{(F-1)}$ states, while semiclassical first-law/Hawking-temperature matching assigns one independent boundary qubit the area increment $\Delta A=4 \ln(2) l_P^2$. The resulting discrete horizon tower admits an explicit energy-conserving unitary evaporation map, yields the state-counting detailed-balance factor $2^{(-q)}$, and produces a Page turnover at $n=n_0/2$; in the macroscopic Schwarzschild limit $M_{\text{Page}}=M_0/\sqrt{2}$ and the simplified evaporation-time fraction is $1-1/(2 \sqrt{2})$. Causality and collective horizon updating give the Hawking mass scalings Γ proportional to $M^{(-1)}$, P proportional to $M^{(-2)}$, and lifetime proportional to M^3 . A horizon-blindness argument further removes the need for an independent phenomenological horizon coupling: in a universal strict-local QCA the event horizon cannot carry a separate surface gate that is not encoded in local records. Microcanonical reciprocity then relates the inclusive spontaneous-emission strength to the corresponding absorption probability by the boundary-state degeneracy ratio. The remaining tasks are computational and structural rather than a free source normalization: calculate the QCA greybody/channel spectrum, construct the full nonlinear geometry/matter QCA, determine exclusive microstate amplitudes and possible level splittings, and derive the boundary-record minimality condition. The framework is therefore a substantially sharpened, but still conditional, finite-information quantum-gravity completion with regular black holes and unitary information transfer.

Keywords: quantum gravity; finite information; finite Weyl algebra; quantum cellular automaton; BCC lattice; emergent spacetime; ADM/DeWitt gravity; regular black holes; horizon entropy; unitary evaporation; Page curve

Claim-status summary

Element	Status	Role
Fixed graph and universal update index	Foundational postulates	Define the microscopic substrate.
Finite local Hilbert space	Additional quantum postulate	Finite information is explicit.
Finite local energy/capacity	Foundational completion postulate	Sets the ultraviolet density scale.
Finite-Weyl algebra	Preferred microscopic carrier	Exact finite unitary record/shift algebra.

Element	Status	Role
Hard projection	Negative-control diagnostic	Shows boundary leakage/operator proliferation.
Exact microscopic update composition	Candidate exact ultraviolet principle	Closure belongs to the finite update/groupoid level.
ADM/DeWitt HDA	Infrared universality target	Recovered in smooth/refinement sectors.
Exact full-cutoff infinitesimal HDA	Rejected in current site realization	Blocked by finite-commutative derivation no-go.
Planck-density core scaling	Robust derived result	Gives r_c proportional to $M^{1/3}$.
Quadratic load law and regular profile	Conditional completion	Requires the stated two-channel/chord-load assumptions.
Horizon-Link entropy	Candidate boundary algebra + IR matching	Rank-2 boundary algebra selects $dL=2$; the $1/4$ coefficient follows conditionally from semiclassical micro-IR matching and boundary-record minimality.
Full many-body Lorentzian completion	Open	Strict-local propagation is explicit at candidate level; the full nonlinear geometry/matter QCA and physical-state construction remain open.

1. Introduction

The central difficulty of quantum gravity is structural. General relativity is not merely a quantum field theory waiting to be placed on a fixed background: its Hamiltonian formulation is constrained, the spatial metric is dynamical, and the brackets of the Hamiltonian and spatial-diffeomorphism constraints contain the inverse metric as a structure function. A microscopic theory must therefore explain both ultraviolet finiteness and the emergence of the general-relativistic deformation structure.

The present framework reverses the usual order of construction. It does not begin by quantizing a continuum metric. It begins with a fixed graph of adjacency relations and a universal ordering of microscopic updates. Proper time, physical distance, smooth geometry, and continuum constraints are collective descriptions of a finite-information substrate. The theory then asks four linked questions: what finite quantum algebra should live on the graph; what notion of closure should be exact; how smooth general relativity can emerge; and what happens when local gravitational load approaches the microscopic capacity.

finite information $\rightarrow$ *finite quantum update algebra* $\rightarrow$ *emergent geometry* $\rightarrow$ *ADM/DeWitt gravity* $\rightarrow$ *regular*

The remainder of this paper develops that chain in detail. Where an assumption is additional rather than derived, it is identified explicitly. Numerical results are treated as pilot consistency tests rather than as proofs.

PART I. MICROSCOPIC FINITE-INFORMATION ARCHITECTURE

2. Capacity, information, and logical status

Finite energy capacity

$$E_{cell} \leq E_p$$

$$\rho_p = \frac{m_p}{\ell_p^3}$$

The energy-capacity postulate fixes the maximum local energy or density scale once a microscopic cell realization is specified. It motivates a Planck-density central cap but does not mathematically imply a finite-dimensional Hilbert space.

Finite Local Information Postulate

$$dim(H_{cell}) = d < \infty$$

Finite local information is therefore retained as an additional explicit quantum postulate. A bounded Hamiltonian may exist on an infinite-dimensional Hilbert space, so finite energy and finite information must remain logically distinct assumptions.

Bounded observables and records

$$\eta = H_{cell}/E_p, 0 \leq \eta \leq 1$$

Physical storage, geometry, and Link observables are represented by bounded Hermitian functions of finite records. The finite-Weyl shift itself is not interpreted as an energy-raising operator; its cyclicity is an information-space property.

3. Diagnostic ultraviolet model: truncation and hard projection

Finite-edge diagnostic

$$[a, a^\dagger] = I - d\Pi_{sat}, \Pi_{sat} = |d-1\rangle\langle d-1|$$

A truncated oscillator demonstrates that exact Heisenberg commutation cannot hold globally in finite dimension. Low states can approximate the ordinary oscillator, while the spectral edge necessarily deforms it. This remains a useful diagnostic but not the preferred microscopic GCC algebra.

Exact projected-commutator identity

$$\begin{aligned} P &= P_{cap}, Q = I - P, H^-[N] = PH[N]P \\ [PAP, PBP] &= P[A, B]P - PAQB P + PBQAP \\ \Delta_{cap}[N, M] &= PH[M]QH[N]P - PH[N]QH[M]P \end{aligned}$$

The identity is exact and exposes capacity-boundary leakage. Exploratory closure audits showed, however, that repeated commutators can generate a large saturation-conditioned operator sector. Hard projection is therefore retained as a negative-control completion rather than the preferred ultraviolet architecture.

4. Preferred microscopic carrier: finite Weyl and symplectic updates

Finite Weyl algebra

$$\begin{aligned} Z|n\rangle &= \omega^n |n\rangle \\ Xe_n &= e_{(n+1) \bmod d} \\ \omega &= \exp\left(\frac{2\pi i}{d}\right) \\ ZX &= \omega XZ \end{aligned}$$

The pair Z,X is an exact finite unitary clock/shift algebra. A phase-symmetrized Weyl basis W(u) closes under multiplication, and its commutator has bounded trigonometric structure coefficients. For low phase-space labels the sine factor is linear and ordinary canonical brackets are recovered as a smooth approximation.

Finite symplectic/Clifford sector

$$G_{micro} \sim HW_d^N \rtimes Sp(2N, Z_d)$$

Finite symplectic transformations preserve the Weyl symplectic product and are implemented by finite Clifford unitaries in the standard admissible cases. This gives an exactly closed finite update carrier. The notation is architectural: details depend on whether d is prime, prime-power, or a more general ring-valued register.

Compute-Link interpretation

Local phase operations represent Compute/record refreshes, while relative Weyl operators such as $Z_i Z_j^\dagger$ represent Link records between neighboring cells. Because these are finite unitaries from the outset, no spectral-edge projector is required to define the microscopic state space.

5. Compute-Link quantum factorization and correlations

$$H_{cell} = H_C \otimes H_L$$

$$\chi_C = \text{Tr}(\rho P_C), \chi_L = \text{Tr}(\rho P_L), q_{CL} = \text{Tr}(\rho P_C P_L)$$

$$\rho/\rho_P = q_{CL}$$

For an uncorrelated Compute-Link state, $q_{CL} = \chi_C \chi_L$. The minimum isotropic closure $\chi_C = \chi_L = \chi$ therefore gives

$$\rho/\rho_P = \chi^2$$

For a correlated state define

$$C_{CL} = \langle P_C P_L \rangle - \langle P_C \rangle \langle P_L \rangle$$

$$\frac{\rho}{\rho_P} = \chi^2 + C_{CL}$$

Thus the quadratic regular-core law is the uncorrelated semiclassical limit of a two-sector quantum architecture. The new Weyl Link construction below supplies a second, complementary interpretation of the same quadratic response.

5.1 Explicit reversible Compute-Link correlation circuit

The covariance term C_{CL} need not be left as an arbitrary radial function. A minimal reversible dilation uses effective two-level Compute and Link records C and L , a memory register R that stores the previous Link state, and a scheduler qubit S . Begin from ρ_{χ^C} tensor ρ_{χ^L} with $\rho_{\chi} = (1-\chi)|0\rangle\langle 0| + \chi|1\rangle\langle 1|$. Let the primitive first-harmonic Link phase be $\phi = 2g\chi^2$, where $g = t_P \epsilon_L / \hbar$. A scheduler rotation $H D(\phi) H$ gives the correlated-branch probability

$$\lambda(\chi) = \sin^2(g\chi^2).$$

On the correlated branch apply $\text{CNOT}_{(C \rightarrow L)}$ followed by $\text{SWAP}_{(L,R)}$. After tracing the reversible work registers R and S , the Compute-Link support becomes

$$q_{CL} = (1-\lambda)\chi^2 + \lambda\chi,$$

$$C_{CL} = \chi(1-\chi) \sin^2(g\chi^2),$$

$$\rho/\rho_P = f_g(\chi) = \chi^2 + \chi(1-\chi) \sin^2(g\chi^2).$$

The correction is bounded, vanishes at $\chi=0$ and $\chi=1$, and begins at order χ^5 in the weak-load expansion. Consequently it does not alter the leading r^{-4} exterior correction of the $p=2$ branch; it contributes only at much higher inverse powers of radius in the weak field. The Link capacity budget implies $g \leq f_L/z$ when a fraction f_L of the cell capacity is assigned to Link processing. This circuit is a concrete minimal dilation, not a uniqueness theorem, but it replaces an arbitrary $C_{CL}(r)$ interpolation by an explicit finite unitary candidate.

6. Quantum metric records and dynamic geometry

Bounded metric record

$$h^{ab} = (EE^T)^{ab}$$

A finite geometry register may encode an inverse-triad-like record E and define the metric through a positive construction $h = E E^T$. Quantum superpositions of finite record labels then produce operator-valued metric records without requiring the microscopic coordinate labels themselves to be continuous.

Conjugate geometry refresh

Exact finite-dimensional canonical commutation is impossible globally. The conjugate geometry refresh is therefore represented by a finite Weyl shift/phase pair. In a low-phase chart the finite pair approximates canonical variables, while the exact microscopic evolution remains finite and unitary.

Geometry-controlled update

$$U_N = \sum_g |g\rangle\langle g| \otimes U_N[g]$$

A finite geometry state can control the local symplectic/Clifford scheduler. Geometry and matter then entangle under a unitary update. In a semiclassical state sharply peaked on one record, the conditional evolution reduces to the corresponding classical metric sector.

7. Infrared constraint algebra and ADM/DeWitt target

Hypersurface-deformation target

$$[H[N], H[M]]/(i\hbar) \rightarrow D[\omega]$$

$$\omega^a = \hbar^{ab} (N \partial_b M - M \partial_b N)$$

The fixed-graph reconstruction developed here obtains this structure from record redundancy, smooth address gauge, and scheduler non-observability. The finite carrier is required to reproduce that algebra in the low-mode refinement limit.

DeWitt kinetic sector

$$H_\lambda[N] = \int d^3x N \left[\frac{\pi^{ab} \pi_{ab} - \lambda \pi^2}{\sqrt{h}} - \sqrt{h} R^{(3)} \right]$$

In three spatial dimensions the general-relativistic DeWitt value is $\lambda=1/2$ in this convention. Independent refinement audits in the infrared reconstruction found closure roots converging to this value and regulator anomalies decreasing with spatial refinement. The finite-Weyl carrier is not used to alter this infrared target.

Full closure requirement

A viable ultraviolet completion must respect H-H, H-D, and D-D together. Bracket-only deformations are insufficient: a state-dependent scalar factor or a componentwise bounded-momentum substitution can look acceptable in H-H and still generate anomalies in the remaining brackets.

8. Closure by automorphism and linear microscopic selection

Finite-Weyl Closure Principle

Let $A_U(O) = U O U^\dagger$ for a finite unitary U . Then

$$[A_U(A), A_U(B)] = A_U([A, B])$$

Hence a Planck-scale finite-information deformation is anomaly-free at the operator-algebra level if it acts on the complete constraint representation as an automorphism rather than modifying one Hamiltonian term in isolation. This does not by itself create new physics; physical record maps must also be fixed, otherwise a global conjugation can be a basis change.

Physical-record-preserving linear maps

Take finite phase-space labels (q, p) and a general linear symplectic map S . If the equal-K physical record q is not changed by an internal Link refresh, $q' = q$, the symplectic conditions reduce the map to

$$q' = q, p' = p + Cq, C = C^T$$

Fixed-graph locality requires $C_{ij}=0$ for non-neighbors. Uniform-vacuum invariance requires $C \cdot 1 = 0$. These conditions identify C as a weighted graph Laplacian. In a homogeneous isotropic reference sector the minimum form is $C = \kappa L_G$.

Reversible scheduler and discrete wave equation

Combining the record-preserving shear with a finite Fourier/Clifford exchange gives the symplectic recurrence

$$q_{K+1} - 2q_K + q_{K-1} = -\kappa L_G q_K$$

Equivalently, with $\Delta_G = -L_G$, the update is the GCC discrete wave equation. For a Laplacian eigenmode $L_G \phi = \lambda \phi$,

$$\cos(\omega) = 1 - \kappa \lambda / 2$$

In a nearest-neighbor one-dimensional reference chain, $\lambda(k) = 4 \sin^2(k/2)$. With the one-link-per-tick calibration $\kappa = 1$, the principal branch gives $\omega = |k|$ and the real-space solution decomposes into left- and right-moving records $F(n-K) + G(n+K)$. Thus the minimum reversible finite scheduler reproduces exact one-link-per-tick propagation in this benchmark.

9. Nonlinear finite-Weyl Link completion

General local harmonic form

Let the finite record phase at cell i be $\theta_i = 2\pi n_i/d$ and define the relative Link phase $\delta = \theta_i - \theta_j$. Locality, exchange symmetry, invariance under a common phase shift, and a zero-energy uniform record imply that a real Link energy can be expanded as

$$V(\delta) = \sum_{m \geq 1} a_m [1 - \cos(m\delta)]$$

The infrared quadratic coefficient fixes only the combination $\sum m^2 a_m$. Therefore finite information, locality, and the infrared wave equation alone do not uniquely select a nonlinear ultraviolet interaction.

Weyl chord load

Define the normalized chord load of two unit-circle Weyl records by

$$\chi_W = |\sin(\delta/2)|, 0 \leq \chi_W \leq 1$$

This is a completion dictionary between relative finite-Weyl geometry and a dimensionless local Link load. It is geometrically natural but is not derived from the original GCC postulates. For the primitive first harmonic,

$$1 - \cos(\delta) = 2\chi_W^2$$

so the nonlinear periodic interaction is exactly quadratic in the chord load even at saturation.

Positive-harmonic uniqueness proposition

Assume $a_m \geq 0$ for each harmonic, normalize the fully separated record $\delta = \pi$ to the saturation scale, and define $f(\chi_W) = V(\delta)/V(\pi)$. Since

$$V(\pi) = 2 \sum_{m \text{ odd}} a_m$$

and for $\chi_W \rightarrow 0$, δ approximately $2\chi_W$,

$$f(\chi_W) = \chi_W^2 \frac{\sum m^2 a_m}{\sum_{m \text{ odd}} a_m} + O(\chi_W^4)$$

Exact infrared normalization $f(\chi_W) = \chi_W^{2+O(\chi_W^4)}$ therefore requires

$$\sum_m m^2 a_m = \sum_{m \text{ odd}} a_m$$

For nonnegative coefficients the difference between the two sides is a sum of nonnegative terms: all even harmonics contribute $m^2 a_m$ and all odd harmonics $m > 1$ contribute $(m^2 - 1)a_m$. Equality therefore forces every coefficient except a_1 to vanish. Within this explicitly stated positive-harmonic sector, the first harmonic is unique up to the overall energy scale:

$$V_{\text{link}}(\delta) = \epsilon_L [1 - \cos(\delta)] = 2\epsilon_L \chi_W^2$$

The proposition fails to be unique if negative harmonic coefficients are permitted and fine-tuned cancellations are allowed. Positivity is therefore a substantive completion condition, not a mathematical consequence of finite Weyl kinematics alone.

Bounded Link response

$$0 \leq V_{link} \leq 2\epsilon_L$$

The Link energy and its finite phase kick are bounded. This realizes finite transfer response without a hard projector. If z is the graph coordination number and Link energy is split between incident cells, the Link share of the cell-capacity budget satisfies $z \epsilon_L \leq EP$, or $z \epsilon_L = f_L EP$ if only a fraction f_L of the cell capacity is assigned to Link processing.

PART II. MICROSCOPIC CLOSURE AND EMERGENT GENERAL RELATIVITY

10. Why exact finite-cutoff infinitesimal HDA is too strong

A finite-commutative derivation no-go

Let $A = C^N$ be the algebra of complex functions on N graph sites with pointwise multiplication. Suppose $\delta: A \rightarrow A$ is a derivation satisfying the ordinary Leibniz rule $\delta(fg) = \delta(f)g + f\delta(g)$. Let e_i be the indicator function of site i . Since $e_i^2 = e_i$,

$$\delta(e_i) = \delta(e_i^2) = 2e_i\delta(e_i)$$

At component i this gives $\delta(e_i)_i = 2\delta(e_i)_i$, hence $\delta(e_i)_i = 0$. At every component $j \neq i$, the right-hand side vanishes, so $\delta(e_i)_j = 0$. Therefore $\delta(e_i) = 0$ for every basis idempotent and

$$\delta = 0.$$

Thus a finite commutative algebra of site values admits no nonzero derivation with the ordinary continuum Leibniz property. This does not forbid finite difference operators, noncommutative differential calculi, or exact finite group actions. It forbids identifying an ordinary continuum spatial derivative with a nontrivial exact derivation of the finite pointwise site algebra.

Consequence for the closure target

The conclusion is not that discrete gravity cannot possess exact symmetry. Rather, exactness should be formulated at the level appropriate to the microscopic variables: finite transformations, noncommutative record algebras, groupoids, or specially constructed perfect discretizations. The continuum HDA can then be a derived low-energy algebra. This distinction parallels the general observation that discretization typically breaks continuum diffeomorphism symmetry, while specially improved or perfect constructions can restore exact symmetry in suitable sectors.

11. Exact microscopic Weyl composition

To verify that finite microscopic exactness is not merely formal, Eq. (3.1) was evaluated as an explicit matrix identity. For each d , X and Z were constructed as $d \times d$ unitary matrices and the relative Frobenius residual was computed:

$$R_W = \frac{\|W_M W_N - \omega^k W_N W_M\|_F}{\|W_M W_N\|_F}$$

The residuals are at floating-point roundoff. The finite Weyl sector therefore provides a concrete exactly closed microscopic operator algebra at finite Hilbert-space dimension.

12. Geometry-derived normal-normal bracket

Continuum geometric identity

Consider a spatial hypersurface with embedding $X(s^a)$, tangent vectors $e_a = \partial_a X$, induced metric $h_{ab} = e_a \cdot e_b$, and unit normal n . A normal deformation with lapse N is

$$\delta_N X = Nn.$$

Varying $n \cdot e_a = 0$ and $n \cdot n = 1$ gives a purely tangential normal variation. Since $\delta_N n \cdot e_a = \partial_a (N n) \cdot e_a$,

$$(\delta_N n) \cdot e_a = -\partial_a N$$

Writing $\delta_N n = C^a e_a$ and solving with h_{ab} yields

$$\delta_N n = -h^{ab}(\partial_b N)e_a.$$

The order difference of two normal deformations is therefore

$$[\delta_N, \delta_M]X = e_a h^{ab}(N\partial_b M - M\partial_b N).$$

Equation (6.3) is the H-H hypersurface-deformation structure function. Importantly, h^{ab} arises from solving the orthogonality condition for the updated normal; it need not be inserted independently into the primitive normal-refresh rule.

Fixed-graph discretization

On a graph refinement of a smooth patch, tangent records are reconstructed locally from neighboring embedding records. A minimal centered realization is $e_a(i)$ approximately $[X(i+a) - X(i-a)]/(2a)$. The normal is then obtained as the normalized vector orthogonal to the three local tangents. After the first normal refresh, all tangents and normals are recomputed from the updated records before the second refresh. This state dependence is essential; reusing the original normal would remove the metric-dependent response.

Three-dimensional non-orthogonal pilot audit

A reproducible pilot calculation used a 15^3 graph on $[-1,1]^3$ embedded in R^4 as $X=(u,v,w,F)$, with

$$F = 0.18uv + 0.11uw - 0.14vw + 0.07u^2 - 0.05v^2 + 0.04\sin(\pi u)\sin(\pi w)$$

$$N = 0.75 + 0.12u - 0.08v + 0.06w + 0.04uv$$

$$M = 0.65 - 0.09u + 0.11v + 0.05w - 0.03vw$$

For each epsilon, the two finite paths X_{NM} and X_{MN} were computed with the normal recalculated after the first step. The order difference was fitted to ϵ^2 times the vector predicted by Eq. (6.3). Two outer graph layers were excluded from the norm.

A log-log fit gives raw scaling $\epsilon^{1.999}$ and residual scaling $\epsilon^{3.084}$. At $\epsilon=0.01$, direct use of the full inverse metric gives a relative vector residual of 2.59×10^{-3} , whereas replacing h^{ab} by δ^{ab} increases the residual to 3.62×10^{-2} . This is a pilot consistency check, not a proof for arbitrary graphs, but it verifies the required order structure and the necessity of the inverse-metric response in a non-orthogonal three-dimensional example.

13. Full infrared hypersurface-deformation target

In the smooth limit the effective normal and tangential generators are required to satisfy the standard geometric deformation relations, up to conventional signs:

$$[\delta_N, \delta_M] = \delta_v v^a = h^{ab}(N\partial_b M - M\partial_b N)$$

$$[\delta_v, \delta_N] = \delta_{L_v N}$$

$$[\delta_v, \delta_w] = \delta_{[v,w]}$$

At the canonical level these become the H-H, D-H, and D-D sectors of the ADM/DeWitt constraint algebra. The ADM/DeWitt theory is the required infrared universality class. The remaining question is how the finite graph approaches this target and where exactness can consistently be imposed.

14. Finite-difference closure audit: full cutoff versus smooth sector

Kinematic test operators

To isolate the effect of finite spatial differentiation, consider a periodic one-dimensional graph with centered derivative D_a . For a site lapse N and shift v define the multiplication operator M_N and tangential transport $L_v = M_v D_a$. The continuum identities suggest

$$\begin{aligned} [L_v, M_N] &= M_{v \partial N} \\ [L_v, L_w] &= L_{v \partial w - w \partial v} \end{aligned}$$

At finite spacing we compare the left-hand sides with the same expressions using D_a . The test profiles were smooth trigonometric functions, and low-energy restriction used fixed Fourier modes $|k| \leq 3$ as the graph was refined.

Full-cutoff operator residual

The full matrix residuals do not decrease toward zero because the highest graph modes remain at the cutoff and the centered difference operator is not an exact derivation of the finite pointwise algebra. This numerical behavior is the finite-difference manifestation of the no-go result in Section 4.

Fixed smooth-sector residual

With the low Fourier sector held fixed while the graph is refined, the residuals scale as

$$A_{HD}^{low} \propto a^{1.987}, \quad A_{DD}^{low} \propto a^{1.981}$$

If both input and output are projected back to the same low sector, the measured exponents are 1.995 and 1.999 respectively. Thus the ordinary infinitesimal deformation algebra is recovered with the expected second-order accuracy of the centered derivative in the smooth sector, even though it is not an exact identity on the complete finite-cutoff operator space.

15. From geometry response to a finite-Weyl scheduler

Equation (6.2) suggests a microscopic dictionary that does not hard-code the continuum HDA. Let the primitive geometry update determine a finite record b_N^a from the measured tangential response of the normal. In a low-load chart b_N^a approaches $h^{ab} \partial_b N$. Quantize the lapse amplitude and this response into finite labels A_N and B_N^a and use them as Weyl exponents.

$$W_N^a = X^{A_N} Z^{B_N^a}$$

The order dependence of two such updates is controlled exactly by the finite symplectic product

$$\kappa^a = A_N B_M^a - A_M B_N^a.$$

After conversion back to a smooth chart, Eq. (9.1) approaches

$$\kappa^a / (s_A s_B) \rightarrow N b_M^a - M b_N^a \rightarrow h^{ab} (N \partial_b M - M \partial_b N)$$

This construction separates two questions. The Weyl composition is exact at finite d . The identification of its finite labels with continuum lapse/normal-response variables is an emergent chart whose accuracy is controlled by graph and record refinement.

16. Spatial address and internal frame gauge

Address relabeling

In the fixed-graph interpretation, a tangential deformation is not a literal motion of the fundamental graph. It is an update of relational address records or a graph relabeling in the smooth description. Therefore the compensating finite transformation associated with κ^a belongs naturally to an address-gauge sector rather than to a new propagating degree of freedom.

Triad redundancy

$$h = EE^T, E \rightarrow ER, R \in SO(3)$$

A local triad record contains a frame redundancy invisible to the metric. This supplies an independent internal gauge sector. Finite $SO(3)$ frame transformations can be lifted to finite-dimensional $SU(2)$ representations. The present paper does not claim that an explicit Gauss term must appear in every H-H bracket; that depends on variables and gauge choice. The key requirement is that internal frame changes remain within the allowed gauge fiber rather than appearing as an anomaly outside the physical record algebra.

17. Two-level closure principle

The combined results motivate the following GCC closure principle.

Microscopic level: $A_{micro} = 0$ for exact finite update/groupoid composition

Emergent level: $A_{eff} \rightarrow 0$ in the smooth low – phase refinement sector

These statements are logically different. The first concerns exact finite quantum consistency. The second concerns recovery of the continuum first-class constraint algebra. Demanding the second as an exact identity on the entire finite-cutoff commuting site algebra is neither necessary for emergence nor compatible with the derivation no-go of Section 4.

This viewpoint also clarifies the role of unitary automorphisms. If $A_U(O) = U O U^\dagger$, then $[A_U(A), A_U(B)] = A_U([A, B])$, so a complete conjugation preserves an already closed algebra. But it maps any existing anomaly by the same conjugation and therefore does not by itself solve closure. This framework instead places exactness in the primitive finite update law.

18. ADM/DeWitt infrared limit

In the infrared, GCC requires canonical metric variables to approach the ADM/DeWitt form. In three spatial dimensions the Hamiltonian density is, in standard conventions,

$$H = (1/\sqrt{h}) \left[\pi^{ab} \pi_{ab} - (1/2) \pi^2 \right] - \sqrt{h} R$$

$$H_a = -2D_b \pi_a^b$$

The finite-Weyl composition law alone does not determine the DeWitt kinetic coefficient. The infrared reconstruction separately selects the ADM/DeWitt universality class. The result established here is compatibility of a finite exactly closed microscopic carrier with the geometric structure function and with the full smooth HDA target.

18.1 Reproducible numerical closure-audit tables

d	Weyl phase exponent	Relative matrix residual
5	2	2.18×10^{-16}
7	4	5.75×10^{-16}
11	8	3.54×10^{-16}
17	14	4.31×10^{-16}

epsilon	Raw normal-normal diamond	Residual after fitted tangential shift	Fitted coefficient
0.040	3.17×10^{-4}	2.52×10^{-6}	0.99552
0.020	7.93×10^{-5}	3.01×10^{-7}	0.99723
0.010	1.98×10^{-5}	3.49×10^{-8}	0.99810

epsilon	Raw normal-normal diamond	Residual after fitted tangential shift	Fitted coefficient
0.005	4.96×10^{-6}	4.16×10^{-9}	0.99853

Sites	a	Full-cutoff H-D residual	Full-cutoff D-D residual
32	0.19635	1.2288	1.1354
64	0.09817	1.2257	1.1224
128	0.04909	1.2250	1.1191
256	0.02454	1.2248	1.1183

Sites	a	Low-mode H-D residual	Low-mode D-D residual
32	0.19635	0.11249	0.17904
64	0.09817	0.02875	0.04623
128	0.04909	0.007227	0.01165
256	0.02454	0.001809	0.002918

18.2 Strict-local BCC QCA and processing-scale audit

Primitive QCA rather than a finite-time local Hamiltonian

If strict one-neighborhood-per-K causality is fundamental, the primitive evolution should be a finite-radius unitary QCA rather than $U = \exp(-i t_* H_{\text{local}}/\hbar)$. Even a nearest-neighbor H_{local} generates $H_{\text{local}}^2, H_{\text{local}}^3, \dots$ in the exponential and therefore generally produces arbitrarily distant small-amplitude tails at finite time. In the present branch U_{QCA} is primitive, while $H_{\text{eff}} = (i \hbar/t_*) \text{Log } U_{\text{QCA}}$ is only an infrared effective generator.

Minimal BCC carrier

A strict-local candidate is obtained on the eight-neighbor BCC graph with a doubled Clifford carrier. Let $\alpha_i = \sigma_x \otimes \sigma_i$ and $P_i^{(+/-)} = (I \pm \alpha_i)/2$. For $s = (s_x, s_y, s_z)$, $s_i = \pm 1$, define $A_s = P_x^{(s_x)} P_y^{(s_y)} P_z^{(s_z)}$. The one-tick law is

$$\psi_{(K+1)}(n) = \sum_s A_s \psi_K(n-s).$$

This is a single radius-1 convolution, not three physically sequential substeps. Its support therefore advances by at most one graph neighborhood per K. The long-wavelength branch is isotropic with $c_{\text{IR}} = a/t_*$, and the BCC construction saturates the three-dimensional isotropic speed bound of this directional class. The four-component carrier contains two positive-frequency amplitudes and is the minimum doubled Clifford realization used here for the two physical TT gravitational-wave polarizations.

Published classifications of homogeneous, local, isotropic and unitary two-component quantum walks on $\mathbb{Z}^3$ find a unique admissible Cayley-graph geometry in three dimensions and two Weyl walks, corresponding to the two chiralities [10,11]. The present $d=4$ propagation carrier can be viewed as a doubled-Weyl structure. This supports BCC as the preferred minimal linear propagation graph; it is not a theorem excluding all higher-dimensional, nonlinear or non-Abelian QCA completions.

Relational processing length and capacity consistency

The graph-edge length is not identified directly with the observed continuum distance. Define instead the relational second-moment processing length

$$\text{ell}_{\text{proc}}^2 = (1/(3z)) \sum_s h_{\text{ab}} d_s^a d_s^b.$$

For the isotropic BCC neighbor shell, $\text{ell}_{\text{proc}} = a$ and the physical node volume in the auxiliary orthonormal realization is $V_{\text{node}} = 4 \text{ell}_{\text{proc}}^3$. The static graph Laplacian and the dynamical infrared cone are then normalized by the same relational scale.

Finite energy capacity and finite information dimension must remain separate. In particular, $d=4$ does not imply a four-times-larger energy capacity. The earlier Minimal Action Saturation identification $\zeta = d$ is therefore not adopted here as a derived law. If one identifies one BCC processing node with one GCC capacity cell whose maximum energy is E_P and whose saturation density is ρ_P , consistency gives

$$4 \text{ell}_{\text{proc}}^3 = l_P^3,$$

$$\text{ell}_{\text{proc}} = 4^{(-1/3)} l_P, \quad t_* = 4^{(-1/3)} t_P.$$

This is a conditional node=cell calibration branch, not a replacement for the foundational capacity postulate. If a QCA node instead coarse-grains several capacity cells, the numerical processing length changes while the distinction between energy capacity and information dimension remains mandatory.

PART III. FINITE CAPACITY AND REGULAR STRONG GRAVITY

19. Fundamental cell capacity

Let one elementary grid cell possess a maximum energy capacity equal to the Planck energy:

$$E_{cap} = E_p = m_p c^2.$$

The elementary reference volume is

$$V_p = \ell_p^3.$$

Hence the maximum mass-density scale is

$$\rho_p = m_p / \ell_p^3 = c^5 / (\hbar G^2).$$

The physical interpretation is that no elementary cell is required to represent arbitrarily large localized energy content. Define the cell occupancy fraction

$$\eta_i = E_i / E_p, 0 \leq \eta_i \leq 1.$$

The storage-saturation condition $\eta_i \rightarrow 1$ should be distinguished from processing or Link saturation associated with lapse suppression. A black-hole horizon need not consist of Planck-density cells. In the present completion, the finite-density core and the horizon-scale scheduler response remain logically distinct.

20. Core radius from capacity alone

Consider a spherically symmetric object of total mass M . If the central region reaches the maximum density ρ_p , the smallest volume capable of storing the mass is $V_c = M / \rho_p$. For a spherical core,

$$(4\pi/3)r_c^3 = M / \rho_p.$$

Therefore

$$r_c = [3M / (4\pi\rho_p)]^{1/3}.$$

Using Planck quantities,

$$r_c = \ell_p [3M / (4\pi m_p)]^{1/3}.$$

Thus

$$r_c \propto M^{1/3}.$$

This scaling does not depend on a Hayward metric, on the two-channel rule, or on the finite-Weyl completion. It follows directly from finite Planck cell capacity, three-dimensional volume counting, and finite total mass. The numerical coefficient is specific to spherical packing at the saturation density; the $M^{1/3}$ scaling is the robust part of the argument.

21. Local load variable

Let $m(r)$ denote the mass enclosed inside areal radius r . Define the mean enclosed density

$$\bar{\rho}(r) = 3m(r) / (4\pi r^3).$$

The corresponding dimensionless capacity-load variable is

$$\chi(r) = \rho^-(r)/\rho_p = 3m(r)/(4\pi\rho_p r^3).$$

This variable is constructed from the state enclosed by the local spherical region and does not explicitly reference the final asymptotic mass in its local definition. The saturated central limit corresponds to $\chi \rightarrow 1$.

22. Two-channel scheduler completion and microscopic Link bridge

The GCC interpretation distinguishes two operational roles: local state computation (Compute) and transfer/consistency of neighboring state records (Link). Introduce the additional completion hypothesis that stable local M-state support requires both Compute and Link commits.

$$\rho/\rho_p = \chi_C \chi_L.$$

For the minimum spherical and isotropic model, assume that the two channels respond to the same local load variable:

$$\chi_C = \chi_L = \chi.$$

Then

$$\rho(r) = \rho_p \chi(r)^2.$$

Substitution of Eq. (4.2) gives

$$\rho(r) = \rho_p \left[3m(r)/(4\pi\rho_p r^3) \right]^2.$$

Equation (5.3) remains the central completion assumption of the regular-core derivation. It is not derived from the cell-capacity bound alone. Its role is to translate the finite capacity of two required local processing channels into a constitutive relation for the coarse-grained density profile.

Microscopic finite-Weyl bridge

The finite-Weyl Link construction supplies a candidate microscopic realization of the same quadratic response. Let neighboring records carry phases θ_i and θ_j , with relative phase $\delta = \theta_i - \theta_j$. Define the normalized chord load

$$\chi_W = |\sin(\delta/2)|, 0 \leq \chi_W \leq 1.$$

For the primitive first harmonic of a local, exchange-symmetric finite-Weyl Link energy,

$$V_{link}(\delta) = \epsilon_L [1 - \cos(\delta)] = 2\epsilon_L \chi_W^2.$$

If the coarse spherical load is identified with this finite-Weyl chord load,

$$\chi_W(r) = \chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

then the normalized Link response is exactly quadratic at all loads, including saturation. This supplies a microscopic realization of Eq. (5.3), not an independent derivation from the original GCC postulates. The dictionary in Eq. (5.7) is an explicit completion choice.

Conditional first-harmonic selection

A general real, local, shift-invariant and exchange-symmetric periodic Link energy can be written

$$V(\delta) = \sum_{m \geq 1} a_m [1 - \cos(m\delta)].$$

Assume $a_m \geq 0$, normalize the fully separated record $\delta = \pi$ to the saturation scale, and require the infrared normalized response to be exactly χ_W^2 at quadratic order. Because $V(\pi) = 2 \sum_{(m \text{ odd})} a_m$ and δ approximately $2\chi_W$ near the vacuum, the infrared coefficient condition is

$$\sum_m m^2 a_m = \sum_{m \text{ odd}} a_m.$$

For nonnegative coefficients, every even harmonic contributes positively only to the left side, while every odd harmonic with $m > 1$ contributes the positive excess $(m^2 - 1)a_m$. Equality therefore forces $a_m = 0$ for all $m \neq 1$. Within this stated positive-harmonic sector, the first cosine harmonic is unique up to its overall energy scale. Negative coefficients or more general microscopic interactions can evade this proposition, so full microscopic uniqueness remains open.

23. Analytic regular mass profile

Spherical mass conservation gives

$$dm/dr = 4\pi r^2 \rho(r).$$

Using Eq. (5.4),

$$dm/dr = 9m(r)^2 / [4\pi \rho_p r^4].$$

Equivalently,

$$d[1/m(r)]/dr = -9/[4\pi \rho_p r^4].$$

Integration yields

$$1/m(r) = C + 3/[4\pi \rho_p r^3].$$

Imposing $m(r \rightarrow \infty) = M$ fixes $C = 1/M$, so

$$m(r) = Mr^3 / [r^3 + 3M/(4\pi \rho_p)].$$

Defining

$$a^3 = \frac{3M}{4\pi \rho_p}$$

the result takes the compact form

$$m(r) = Mr^3 / (r^3 + a^3).$$

This is a Hayward-type regular-black-hole mass profile. It appears here as the analytic consequence of the $p = 2$ local finite-capacity closure. The finite-Weyl Link construction supplies a conditional microscopic realization of the same closure.

24. Central and exterior limits

Central limit

For $r \ll a$, Eq. (6.7) gives

$$m(r) \simeq (4\pi/3)\rho_p r^3.$$

Therefore the central density is finite and saturates at the capacity scale:

$$\rho(0) = \rho_p.$$

In the standard static spherical line element $f(r) = 1 - 2Gm(r)/(c^2 r)$, the central behavior is de Sitter-like rather than singular.

Exterior limit

For $r \gg a$,

$$m(r) = M[1 - a^3/r^3 + o(r^{-6})].$$

Consequently the leading correction to the Schwarzschild metric function is

$$\Delta f(r) \propto r^{-4}.$$

The correction is strongly localized toward the deep interior and strong-field region, while the far exterior rapidly approaches general relativity.

25. Fixing the strong-gravity length scale

In geometrized units $G = c = 1$, write the regular profile in the form

$$m(r) = Mr^3 / (r^3 + 2M\ell_*^2).$$

The geometrized Planck density is

$$\rho_p^{(geom)} = G\rho_p/c^2 = 1/\ell_p^2.$$

Equation (6.6) therefore becomes

$$a^3 = (3/4\pi)M\ell_p^2.$$

Equating the denominators of Eqs. (6.7) and (8.1),

$$2M\ell_*^2 = (3/4\pi)M\ell_p^2.$$

Hence

$$\ell_*^2 = [3/(8\pi)]\ell_p^2$$

$$\ell_* = \sqrt{3/(8\pi)}\ell_p \simeq 0.3455\ell_p.$$

Under the $p = 2$ completion, the regularization scale is fixed by the Planck length rather than fitted independently for each compact object.

26. General p-channel completion and harmonic robustness

To test whether the exponent two is unique at the effective level, consider the generalized local closure

$$\rho/\rho_p = \chi^p.$$

For $p > 1$, the spherical mass equation can be integrated exactly. Using $m(\text{infinity}) = M$ gives

$$m(r) = M1 + \left[(a^3/r^3)^{p-1} \right]^{-1/(p-1)}$$

$$a^3 = 3M/(4\pi\rho_p).$$

All $p > 1$ solutions share the same Planck-density central limit,

$$m(r) \simeq (4\pi/3)\rho_p r^3 (r \rightarrow 0).$$

Thus Planck-density saturation by itself does not select $p = 2$.

Why $p = 1$ fails for an isolated finite mass

For $p = 1$, Eq. (9.1) gives $\rho = \bar{\rho}$. The mass equation reduces to

$$dm/dr = 3m/r.$$

Its nonzero solution is m proportional to r^3 and therefore cannot approach a finite nonzero ADM mass at large radius. Within this local closure, an asymptotically finite isolated object requires $p > 1$. If p counts an integer number of multiplicative processing channels, the smallest admissible value is $p = 2$.

Relation of $p = 2$ to the finite-Weyl completion

The finite-Weyl result of Section 5 does not make the effective p -channel argument redundant; it supplies a second, conditional route to the same quadratic response. The effective argument says that $p = 2$ is the smallest integer multiplicative closure compatible with finite isolated mass. The microscopic argument says that, if the finite-Weyl chord load is identified with χ and one restricts to nonnegative harmonics with exact saturation and infrared normalization, the first harmonic is selected and its energy is exactly proportional to χW^2 . These are complementary rather than independent proofs. Full uniqueness remains open because the load dictionary, positive-harmonic restriction, and microscopic coarse-graining are additional assumptions.

Robustness to a second Weyl harmonic

As a controlled stress test, let

$$V(\delta) = a_1[1 - \cos(\delta)] + a_2[1 - \cos(2\delta)], \epsilon = a_2/a_1.$$

With $\chi = \sin(\delta/2)$, the normalized constitutive law becomes

$$\rho/\rho_p = f_\epsilon(\chi) = \chi^2[1 + 4\epsilon(1 - \chi^2)].$$

The saturated center remains exact because $f_\epsilon(1)=1$. Positivity of the two-harmonic potential requires $\epsilon \geq -1/4$. Linear stability of the regular $\chi = 1$ central branch gives $\epsilon < 1/8$. Thus this simple family admits the consistency interval

$$-1/4 \leq \epsilon < 1/8.$$

At low load, $f_\epsilon(\chi) = (1+4\epsilon)\chi^2 - 4\epsilon\chi^4 + \dots$, so the power of the leading exterior correction remains r^{-4} but its coefficient changes with microscopic harmonic content. The central Planck density and $M^{1/3}$ scaling are therefore more robust than the exterior correction coefficient.

27. Exterior corrections for general p

At large radius, Eq. (9.2) gives

$$m(r) \simeq M - [1/(p-1)](a^3/r^3)^{p-1} + \dots$$

Since $f(r) = 1 - 2m(r)/r$ in geometrized units, the first metric correction scales as

$$\Delta f(r) \propto r^{-(3p-2)}.$$

Therefore

$$p = 2 \Rightarrow \Delta f \propto r^{-4}$$

$$p = 3 \Rightarrow \Delta f \propto r^{-7}.$$

Higher-channel models are progressively more hidden from exterior observations. The $p = 2$ case is simultaneously the minimal finite-mass integer closure and the least suppressed member of this regular family. Within the Weyl stress test, small higher-harmonic admixtures preserve the r^{-4} power while modifying its coefficient.

28. Astrophysical consequences

For the $p = 2$ Planck-capacity completion, the characteristic core radius scales as

$$a \propto (M \ell_p^2)^{1/3}$$

whereas the Schwarzschild radius scales linearly with M . Hence

$$a/r_s \propto M^{-2/3}.$$

For stellar and supermassive black holes this ratio is extraordinarily small. The completion therefore does not naturally predict percent-level departures in ordinary shadow radii, ISCO locations, binary inspiral waveforms, or standard ringdown frequencies. Instead, the natural phenomenology is a general-relativistic exterior together with a finite central core.

The finite-Weyl bridge does not change this conclusion. In the preferred first-harmonic completion it reproduces the same Hayward-type profile. Higher-harmonic deformations can alter the coefficient of the leading r^{-4} correction, but for a Planck-scale core the absolute exterior deviation remains extremely suppressed for astrophysical masses.

29. Horizons, echoes, and information

A regular center does not automatically eliminate an outer event horizon. It therefore does not, by itself, predict gravitational-wave echoes. Echoes require an additional reflecting or horizonless structure capable of returning information to the exterior.

Likewise, removal of the curvature singularity does not by itself solve the black-hole information problem. The information sector therefore has to be constructed independently of the finite core. Part IV develops such a construction: a rank-2 outgoing boundary algebra is extracted from the strict-local $d=4$ carrier, gauge-reduced horizon records are counted, an explicit finite-dimensional unitary evaporation map transfers those records to radiation/history, and a typical-state Page turnover is obtained. These results do not yet calculate the exclusive microscopic amplitudes or the greybody spectrum numerically. Section 35 shows, however, that an independent horizon-source normalization is unnecessary at the inclusive level: microcanonical reciprocity ties spontaneous emission to absorption once the universal local QCA and exterior scattering problem are specified.

30. Finite-Weyl microscopic realization of the quadratic core law

Spherical load dictionary

$$\chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

For spherical strong gravity, define χ as the mean enclosed density in Planck-density units. The microscopic matching condition is

$$\chi_W(r) = \chi(r)$$

This identification is a completion choice. Under it, the positive-harmonic first-link response gives exactly

$$\rho(r)/\rho_p = \chi(r)^2$$

which is the $p = 2$ Compute-Link closure introduced above. The two interpretations are compatible: χ^2 can be read both as the factorized support of two mandatory local sectors and as the squared finite-Weyl chord mismatch of the primitive Link record.

Hayward-type mass profile

$$\frac{dm}{dr} = 4\pi r^2 \rho_p \chi(r)^2$$

$$m(r) = \frac{Mr^3}{r^3 + \frac{3M}{4\pi\rho_p}}$$

$$a^3 = \frac{3M}{4\pi\rho_p}$$

The central density is finite, $\rho(0)=\rho_P$, and the characteristic core scale obeys a proportional to $M^{1/3}$. At large radius $m(r)=M[1-a^{3/r^3}+O(r^{-6})]$, so the corresponding static metric correction begins at order r^{-4} . The finite-Weyl construction therefore supplies a candidate microscopic realization of the closure introduced above as an explicit completion assumption.

Status of the derivation

The chain is conditional. Planck capacity fixes ρ_P and the $M^{1/3}$ scale; finite information and Weyl kinematics provide the finite record carrier; positive-harmonic selection identifies the primitive cosine interaction; and the chord-load dictionary maps that interaction to the coarse spherical load. The dictionary and harmonic positivity are not consequences of $E_{cap}=EP$ alone and must remain separately testable assumptions.

31. Robustness to higher harmonics

Two-harmonic stress test

$$V(\delta) = a_1[1 - \cos(\delta)] + a_2[1 - \cos(2\delta)], \epsilon = a_2/a_1$$

Using $\chi = \sin(\delta/2)$ and normalizing by the first-harmonic saturation value gives

$$f_\epsilon(\chi) = \chi^2[1 + 4\epsilon(1 - \chi^2)]$$

Hence $f_\epsilon(1)=1$: the Planck-density center remains fixed. The spherical autonomous equation is

$$r \frac{d\chi}{dr} = 3[f_\epsilon(\chi) - \chi]$$

Linearization around the saturated center gives a regular inward approach when $\epsilon < 1/8$. Requiring the two-harmonic potential to remain nonnegative for all χ in $[0,1]$ gives $\epsilon \geq -1/4$. Thus the simple two-harmonic model has the regular positive-potential interval

$$-1/4 \leq \epsilon < 1/8$$

Robust versus model-dependent strong-gravity statements

At low load,

$$f_\epsilon(\chi) = (1 + 4\epsilon)\chi^2 - 4\epsilon\chi^4 + O(\chi^6)$$

so the leading exterior power remains r^{-4} while its coefficient changes by the factor $1+4\epsilon$. The Planck-density central limit and the $M^{1/3}$ capacity scaling are therefore more robust than the numerical coefficient of the far-field correction. This distinction is important for phenomenology: the existence and scaling of the core are capacity-level statements, whereas detailed exterior coefficients probe the microscopic harmonic content.

PART IV. BLACK-HOLE INFORMATION AND HORIZON-LINK ENTROPY

32. Why the Planck core cannot store the full black-hole entropy

$$N_{core} \sim M/m_p$$

$$S_{core}^{max} \sim k_B(M/m_p) \ln(d_c) \propto M$$

$$S_{BH} = \frac{k_B A_H}{4\ell_p^2} = 4\pi k_B \left(\frac{M}{m_p}\right)^2 \propto M^2$$

For mass-independent finite local dimension d_C , the Planck-density core has parametrically too few independent states to account for the entropy of a large black hole. The new microscopic Link completion does not change this counting mismatch; it strengthens the distinction between core regularization and horizon-scale information accounting.

33. Horizon boundary algebra and the area law

Why raw graph crossings are not horizon qubits

The information-bearing horizon sector must be defined after physical coarse graining and gauge reduction, not by counting raw BCC edges. For a chosen surface normal $\mathbf{n}$, the raw crossing map $C_n = \sum_{\mathbf{s} \cdot \mathbf{n} > 0} A_{\mathbf{s}}$ can have rank two for special orientations and rank four for generic orientations. Raw crossing-edge number and raw crossing

rank are therefore regulator- and orientation-dependent. This rules out the naive identification “one graph edge crossing the horizon = one independent qubit.”

Orientation-independent normal-current sector

The $d=4$ doubled Clifford carrier has $\alpha_i = \sigma_x$ tensor σ_i . For any unit physical normal n define $\alpha_n = n_i \alpha_i$. The Clifford relations give $\alpha_n^2 = I$ and $\text{Tr}(\alpha_n) = 0$, so its spectrum is $\{+1, +1, -1, -1\}$. The projectors

$$P_{\text{out}}(n) = (I + \alpha_n)/2, \quad P_{\text{in}}(n) = (I - \alpha_n)/2$$

therefore have rank two for every orientation. Under a local frame rotation α_n changes by unitary conjugation, so the rank is gauge/basis independent. The effective outgoing boundary algebra is

$$A_H = P_{\text{out}} M_4(C) P_{\text{out}} \simeq M_2(C).$$

Its minimal finite Weyl algebra is the Pauli/Weyl algebra HW_2 , with a Z_2 central phase label. Thus the strict-local bulk carrier supplies a natural two-state effective crossing sector rather than requiring $d_L=2$ to be inserted by hand.

This kinematic result is not by itself a theorem about the full many-body horizon Hilbert space. The additional Boundary Record Minimality Principle used below states that, after the inside/outside split and gauge reduction, the minimum independent horizon record is precisely this outgoing support algebra, with no extra independent occupancy or species factor. Under that explicit assumption,

$$d_L = 2.$$

Gauge-reduced state counting

Represent F effective horizon patches by binary boundary labels b_i in Z_2 . For a closed connected horizon impose one global Z_2 consistency condition,

$$\sum_{i=1}^F b_i = 0 \pmod{2}.$$

The physical state count is then

$$\Omega_F = 2^{(F-1)},$$

and $S_H = k_B (F-1) \ln 2$. The missing factor of two is an $O(1)$ global constraint and does not change the macroscopic area density. This is the Z_2 specialization of the finite-Weyl boundary-phase construction: a central phase that is record-invisible for a closed bulk update composition becomes a physical edge record when the system is cut by a horizon.

Area normalization by micro-IR matching

The finite-Weyl algebra determines the two-state boundary information increment, $\Delta S = k_B \ln 2$, but it does not by itself determine the physical area carried by that record. The area scale is fixed by the required semiclassical GCC infrared limit. For a stationary horizon,

$$\delta E = (\kappa c^2 / (8 \pi G)) \delta A,$$

$$k_B T_H = \hbar \kappa / (2 \pi c).$$

Therefore $\delta S = \delta E / T_H = k_B \delta A / (4 l_P^2)$. Matching one independent boundary qubit gives

$$\Delta A = 4 \ln 2 l_P^2 \simeq 2.772588722 l_P^2.$$

The Bekenstein-Hawking coefficient is therefore not claimed to follow from finite-Weyl kinematics alone. It follows from a micro-IR matching theorem: the microscopic boundary algebra fixes ΔS , while the independently required semiclassical GR/QFT limit fixes the conversion from information to physical area. The resulting effective patch is not a raw microscopic plaquette.

34. Discrete horizon tower and unitary information transfer

Geometry and information labels

The boundary information labels should not be confused with geometry labels. At fixed horizon level n all binary boundary states share the same macroscopic geometry; geometry changes when the number of independent boundary records changes. The horizon Hilbert space is organized schematically as

$$H_H = \text{direct_sum}_n |g_n\rangle \otimes H_{\text{edge}}^{(n)}, \quad \dim H_{\text{edge}}^{(n)} = 2^n.$$

This removes the earlier temptation to assign a distinct geometry or area eigenvalue to the internal label $b=0$ or 1 .

Discrete area and entropy tower

Using the regular extremal geometry as an information-free offset,

$$A_{\text{ext}} = (9/2) l_P^2, \quad r_{\text{ext}} = 3 l_P / \sqrt{8 \pi}.$$

Define

$$A_n = A_{\text{ext}} + 4 n \ln 2 l_P^2,$$

$$S_n = k_B n \ln 2 = k_B (A_n - A_{\text{ext}}) / (4 l_P^2).$$

Thus the $n=0$ endpoint may retain a finite geometric area while $\dim H_{\text{edge}}^{(0)}=1$. A geometric remnant is therefore not an information remnant: it need not store the information of the progenitor black hole.

Energy levels and finite gaps

For the regular spherical branch with $l_* = \sqrt{3/(8 \pi)} l_P$, the horizon relation can be written

$$M(r_h) = (c^2/G) r_h^3 / [2(r_h^2 - l_*^2)].$$

Set $r_n = \sqrt{A_n/(4 \pi)}$ and $M_n = M(r_n)$. The energy released in a one-record transition is

$$\epsilon_n = (M_n - M_{(n-1)}) c^2.$$

For n much greater than one the Schwarzschild limit gives

$$M_n \simeq m_P \sqrt{n \ln 2 / (4 \pi)},$$

$$\epsilon_n \simeq k_B T_H \ln 2,$$

with the exact finite-step Schwarzschild ratio

$$\epsilon_n / (k_B T_H \ln 2) = 2n-2 \sqrt{n(n-1)} = 1 + 1/(4n) + 1/(8n^2) + O(n^{-3}).$$

Explicit unitary evaporation map

Let b_i in Z_2 denote the independent horizon records. A one-step information-and-energy transfer can be written

$$|g_n; b_1, \dots, b_n\rangle |0\rangle_R$$

$$\rightarrow |g_{(n-1)}; b_1 + b_n, b_2, \dots, b_{(n-1)}\rangle |b_n, \epsilon_n\rangle_R,$$

where addition is modulo two. The map is bijective: b_n is retained in radiation and the previous b_1 is recovered from the updated first record and b_n . It therefore extends to an exact finite-dimensional unitary. Energy conservation is blockwise by construction,

$$M_n c^2 = M_{(n-1)} c^2 + \epsilon_n.$$

Repeated application transfers all progenitor boundary information to radiation/history correlations without requiring the final geometric endpoint to carry a large hidden Hilbert space.

Near-endpoint thermodynamic work term

For the regular branch, applying $dE = T dS$ to the entire static family while holding l_* fixed does not remain exact near the core. The discrepancy vanishes in the Schwarzschild limit and at the cold extremal endpoint and is naturally represented as

$$dE = T dS + dW_{\text{core}}/\text{link}.$$

Accordingly, the constant information-area spacing is interpreted as the boundary information calibration, while the near-core static family can exchange additional work with the core/Link sector. This prevents the regular endpoint from being overinterpreted as an ordinary Schwarzschild thermodynamic state.

35. Hawking detailed balance, evaporation rate, and Page turnover

State-counting origin of the thermal factor

Allow a transition that removes q boundary records. Its exact geometry energy is

$$\epsilon_{(n,q)} = [M_n - M_{(n-q)}] c^2.$$

Because $\Omega_n = 2^n$, microreversibility gives the coarse-grained state-count ratio

$$\Gamma_{\text{emit}}/\Gamma_{\text{absorb}} = \Omega_{(n-q)}/\Omega_n = 2^{-(q)}.$$

For a macroscopic Schwarzschild black hole,

$$\epsilon_{(n,q)} \simeq q k_B T_H \ln 2,$$

so

$$2^{-(q)} \simeq \exp[-\epsilon_{(n,q)}/(k_B T_H)].$$

The Hawking Boltzmann factor is therefore compatible with boundary-state counting rather than being inserted as an independent microscopic probability. At finite n the gap is slightly nonthermal:

$$\epsilon_{(n,q)}/(q k_B T_H \ln 2)$$

$$= (2n/q)[1 - \sqrt{1 - q/n}]$$

$$= 1 + q/(4n) + q^2/(8n^2) + O(n^{-3}).$$

Horizon-blindness and no independent surface coupling

In GCC the primitive update is universal and strict-local: the gate acting on a neighborhood may depend on local geometry and matter records, but it is not supplied with a separate global label stating that the neighborhood lies on an event horizon. This matters because event-horizon membership is a global causal property, while the local curvature at the horizon of a sufficiently massive black hole can be arbitrarily small. A nonzero special surface coupling that survives in that limit would therefore require an additional nonlocal horizon flag and would violate the locality/no-retuning interpretation of the microscopic law. The minimal branch consequently imposes no boundary defect: the same geometry-controlled QCA propagates an ingoing mode across the horizon as in any locally regular region. Reflection and transmission seen from infinity arise from propagation through the exterior geometry, not from an independently tuned horizon membrane. Denote the resulting inclusive absorption probability for a unit-normalized asymptotic channel α by $T_\alpha(\omega)$.

Microcanonical reciprocity and inclusive normalization

Let $M_{(j|i)}^\alpha$ be the transition amplitude for absorption from a black-hole microstate i in $H_{(n-q)}$ plus one incoming quantum α to a microstate j in H_n . Microscopic reversibility equates the modulus of this amplitude with that of the inverse emission channel. Define the microcanonical averages

$$P_{\text{abs}}(n-q \rightarrow n; \alpha) = (1/\Omega_{(n-q)}) \sum_{(i,j)} |M_{(j|i)}^\alpha|^2,$$

$$P_{\text{emit}}(n \rightarrow n-q; \alpha) = (1/\Omega_n) \sum_{(j,i)} |M_{(j|i)}^\alpha|^2.$$

The same matrix-element sum appears in both expressions, so the unknown exclusive amplitudes cancel in the ratio:

$$P_{\text{emit}}/P_{\text{abs}} = \Omega_{(n-q)}/\Omega_n = 2^{(-q)}.$$

For the no-boundary-defect branch P_{abs} is the inclusive QCA absorption/greybody probability T_α at the relevant frequency. Hence

$$P_{\text{emit}}(n \rightarrow n-q; \alpha) = 2^{(-q)} T_\alpha(\omega_{(n,q)}).$$

This is stronger than detailed balance alone: once the absorption problem is calculated, there is no independent horizon-source normalization to fit.

Bose/Fermi occupation and QCA flux normalization

For a general emitted energy $\hbar \omega$ define the black-hole entropy loss $\Delta S_{\text{BH}}(E, \omega) = S(E) - S(E - \hbar \omega) > 0$. Repeated occupation of a bosonic mode gives the geometric series generated by $\exp[-\Delta S_{\text{BH}}/k_B]$, while Pauli exclusion truncates the fermionic occupation. In the smooth continuum normalization the inclusive number flux therefore takes the form

$$d^2 N/(dt d\omega) = (1/(2\pi)) \sum_\alpha T_\alpha(\omega) / \{\exp[\Delta S_{\text{BH}}(E, \omega)/k_B] \text{ minus-or-plus } 1\},$$

with the minus sign for bosons and the plus sign for fermions. In the macroscopic limit ΔS_{BH} approximately $\hbar \omega / T_H$, recovering the usual Hawking greybody form. For the exact QCA the factor $1/(2\pi)$ is replaced, when necessary, by the density-of-states and group-velocity measure derived from the QCA dispersion relation and the physical tick t_* ; this measure is fixed by U_{QCA} rather than by a new emission constant. Detailed balance plus absorption is a standard route to Hawking emission including greybody factors [12-14], and here the state-count ratio supplies its microscopic GCC origin.

Sector-centroid line structure and microstate splitting

The discrete area tower implies discrete transition frequencies,

$$\hbar \omega_{(n,q)} = \epsilon_{(n,q)}.$$

If every microstate inside a fixed n sector were exactly degenerate at the sector mass M_n , the minimum model would predict a line spectrum with a thermal envelope rather than an exactly continuous Planck spectrum. That exact degeneracy is not required by the information counting. A full interacting QCA may instead give $E_{(n,\mu)} = M_n c^2 + \delta E_{(n,\mu)}$, with exponentially many microstate-dependent splittings that preserve the sector count 2^n . The centroid spacing remains approximately $k_B T_H \ln 2$ per information record, while individual transition frequencies can become dense. Resolvable Hawking lines are therefore a conditional prediction of a weak-splitting branch, not a mandatory consequence of the boundary entropy construction.

Exact per-tick emission dilation

For a chosen emission channel $V_{(n,q,\alpha)}$, an exact one-tick unitary can be defined on the paired no-emission/emission subspace as a two-dimensional rotation,

$$|n,\mu;0\rangle \rightarrow \sqrt{1-p}|n,\mu;0\rangle + \sqrt{p} V|n,\mu\rangle,$$

$$V|n,\mu\rangle \rightarrow -\sqrt{p}|n,\mu;0\rangle + \sqrt{1-p} V|n,\mu\rangle.$$

Fresh radiation ancillas over repeated ticks give $p = 1 - \exp(-\Gamma t_*)$. For an exclusive microstate channel the detailed p still requires the full interacting QCA. However, p is not an independent phenomenological horizon parameter after microcanonical averaging: the summed emission probability is fixed by the inverse absorption problem and the horizon-state degeneracy ratio derived below.

Macroscopic inclusive rate and scaling

After the absorption problem is specified, the inclusive power is determined without an independent luminosity coefficient. Writing $\Phi_{\text{QCA},\alpha}(\omega)$ for the asymptotic mode-flux density fixed by the QCA dispersion and physical tick, the microcanonical expression is

$$P(E) = \sum_{\alpha} \int d\omega [\hbar \omega \Phi_{\text{QCA},\alpha}(\omega) T_{\alpha}(\omega)] / \{ \exp[\Delta S_{\text{BH}}(E,\omega)/k_B] \text{ minus-or-plus } 1 \}.$$

In the continuum radial normalization Φ_{QCA} approaches $1/(2\pi)$ per partial-wave channel; the exact finite-QCA expression uses the corresponding group-velocity/density-of-states Jacobian.

For a macroscopic Schwarzschild black hole, T_H is proportional to M^{-1} and the exterior scattering problem depends on frequency through dimensionless combinations such as $GM\omega/c^3$. Changing variables to the corresponding dimensionless frequency therefore gives P proportional to M^{-2} and lifetime proportional to M^3 , in agreement with the standard Hawking/Page scaling. The same exponents also follow independently from GCC causality: a horizon-scale collective update cannot coordinate faster than a time of order r_H/c , so its attempt frequency scales as M^{-1} , while one boundary-record energy scales as $T_H \ln 2$ proportional to M^{-1} . Independent Planck-rate emission from every microscopic patch would instead give the wrong mass scaling and is excluded.

Page turnover

For an initially scrambled n_0 -record horizon, after $k = n_0 - n$ records have been transferred to radiation, the typical pure-state entropy is

$$S_R(n) \simeq k_B \ln 2 \min(n_0 - n, n).$$

The Page turnover occurs at $n=n_0/2$. In the macroscopic area-mass relation n proportional to M^2 , this implies $M_{\text{Page}} = M_0/\sqrt{2}$.

Using the simplified Schwarzschild mass-loss law $M(t)^3 = M_0^3(1-t/\tau)$ gives

$$t_{\text{Page}}/\tau = 1 - 1/(2 \sqrt{2}) \approx 0.6464466.$$

Thus “half the horizon information capacity” does not correspond to half of the physical evaporation time.

Matter coupling and remaining exclusive dynamics

Minimal metric coupling still controls the exclusive geometry-to-matter transition structure. Its schematic variation is

$$\delta S_m = (1/2) \int \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu} d^4x.$$

For free fields $T^{\mu\nu}$ is quadratic, so the exclusive fundamental creation process is generically pair/squeezing-like rather than a literal one-particle lowering operator. A full nonlinear geometry/matter QCA is still required to determine individual microstate amplitudes, species/polarization branching, level splittings and linewidths. However, their common overall normalization is not a separate horizon parameter at the inclusive level: the summed matrix-element strength is already fixed by the inverse absorption probability through microcanonical reciprocity. The remaining greybody factors are therefore scattering observables of the same exterior QCA/geometry, not adjustable Hawking coefficients. The present construction fixes the information bookkeeping, sector energy gaps, inclusive emission/absorption normalization, causal mass scaling and Page turnover while leaving the exclusive microscopic channel decomposition open.

PART V. CLAIMS, LIMITATIONS, AND FALSIFIABILITY

36. Claim-status summary

Element	Status	Meaning
Finite local information	Explicit postulate	Finite-dimensional local quantum records.
Finite Weyl/symplectic carrier	Preferred candidate architecture	Exact finite microscopic unitary composition.
Microscopic closure	Exact at update/groupoid level	The ultraviolet consistency statement is finite, not continuum-infinitesimal.
ADM/DeWitt constraint algebra	Emergent infrared target	Recovered in smooth/refinement sectors; full-cutoff exact HDA is not claimed.
Planck-density regular core	Robust capacity result	The density cap and $M^{1/3}$ core scaling follow before choosing the detailed interpolation.
Quadratic $p = 2$ profile	Conditional completion	Requires the stated Compute-Link/chord-load assumptions.
Horizon boundary entropy coefficient	Conditional micro-IR matching	Rank-2 boundary algebra selects $dL=2$; the $1/4$ coefficient follows from semiclassical micro-IR matching, conditional on Boundary Record Minimality.
Full 3+1 many-body dynamics	Open	A strict-local linear propagation QCA is explicit; a unique full nonlinear geometry/matter update and physical-state construction remain to be derived.

37. Falsifiability and consistency tests

- Infrared coarse graining must recover the complete ADM/DeWitt H-H, H-D, and D-D algebra, not only one bracket.
- The finite microscopic carrier must possess a nonempty physical Hilbert space after all constraints and gauge reductions are imposed.
- Any ultraviolet deformation used as new physics must preserve a first-class physical constraint algebra or be demonstrably pure gauge; isolated term-by-term polymerization is insufficient.

- The record-preserving linear scheduler must remain stable under graph refinement and must not generate uncontrolled macroscopic Lorentz or diffeomorphism violation.
- The Weyl chord-load dictionary must be derived, independently motivated, or empirically constrained; if it fails, the microscopic link to the $p = 2$ regular-core completion is lost.
- The positive-harmonic assumption must be justified by stability or microscopic energy positivity. If negative harmonics are physically required, the first-harmonic uniqueness proposition no longer holds.
- The spherical strong-gravity sector must respect $\rho \leq \rho_P$ and recover the $M^{1/3}$ capacity scaling. Detailed exterior coefficients should be treated as microscopic-model dependent until the harmonic content is fixed.
- The explicit reversible Compute-Link correlation circuit must survive extension to the full finite-dimensional many-body geometry sector; its correlation profile must not be replaced by an arbitrary radial interpolation.
- The rank-2 horizon boundary algebra and Z_2 gauge reduction must survive a full many-body construction. Boundary Record Minimality, rather than a fitted state count, is the remaining microscopic assumption behind $dL=2$.
- The explicit unitary evaporation map must be embedded in the full matter/radiation QCA, preserve energy under the microscopic update law, and transfer progenitor information without a large-information remnant.

38. Strong-gravity limitations and falsification conditions

The present construction does not establish the following:

- a first-principles derivation of the Weyl chord-load dictionary $\chi_W(r) = \chi(r)$;
- microscopic uniqueness beyond the explicitly stated positive-harmonic sector;
- exact equality $\chi_C = \chi_L$ in a generic ultraviolet quantum state;
- dynamical formation of the static profile during collapse;
- inner-horizon stability;
- a rotating Kerr-like completion;
- a completed numerical greybody/channel spectrum, exclusive microstate transition amplitudes and linewidths, and a dynamical proof of endpoint stability;
- a full interacting-field Page curve beyond the typical-state finite-record model;
- a derivation showing that the full nonlinear finite-Weyl scheduler coarse-grains uniquely to the static spherical constitutive law used here.

Most importantly, Planck capacity alone does not uniquely imply the Hayward profile. The Hayward form follows from Planck capacity plus the $p = 2$ local constitutive law. The finite-Weyl Link construction supplies a conditional microscopic realization and a positive-harmonic uniqueness proposition within a restricted sector, but negative harmonic coefficients, alternative load dictionaries, or more general interactions can evade that argument. The capacity-derived density bound and $M^{1/3}$ scaling remain the more robust results.

The completion is falsified as a physical model if its required microscopic dynamics cannot preserve the successful low-energy GR correspondence, if the saturated core is dynamically unstable, or if a consistent rotating/evaporating extension cannot be constructed without violating the finite-capacity and no-retuning principles.

39. Interpretation as a quantum-gravity candidate

The architecture obtained here has three attractive features. First, the microscopic carrier is genuinely finite and unitary. Second, the metric-dependent H-H structure function is generated geometrically from the response of the normal rather than added as an arbitrary ultraviolet coefficient. Third, the failure of exact continuum derivations at finite cutoff is not hidden; it is isolated and converted into a controlled emergence requirement.

These features are not sufficient to establish a complete theory of quantum gravity. The current result is best described as a finite-information quantum-gravity candidate architecture with exact finite microscopic composition, an anomaly-free emergent HDA in the tested smooth sectors, a strict-local BCC propagation candidate, a rank-2 horizon boundary algebra, and an explicit unitary black-hole information-transfer map.

40. Open problems and decisive next tests

- Extend the strict-local BCC $d=4$ propagation QCA to one nonlinear geometry/matter many-body update on the full tensor-product graph Hilbert space.
- Demonstrate the complete 3+1 Lorentzian constraint and physical-state structure, including the role of local frame/Gauss gauge in the chosen microscopic variables.
- Compute the absorption/greybody matrix of the strict-local matter QCA on the regular GCC exterior and use microcanonical reciprocity to obtain the inclusive emission spectrum; derive exclusive pair/squeezing amplitudes from the full nonlinear geometry/matter update.
- Define a controlled refinement/renormalization map connecting finite d and finite graph spacing to the continuum observables.
- Show that the finite-Weyl strong-gravity response, regular core, constraint closure, horizon boundary algebra, and evaporation source arise from one common strict-local microscopic update rule.
- Construct physical states or an appropriate groupoid-averaging analogue and demonstrate unitary relational evolution.
- Derive Boundary Record Minimality or an equivalent gauge-reduction theorem, and determine the microstate level splittings/linewidths. The $dL=2$ boundary algebra, micro-IR area matching, unitary transfer map, microcanonical emission/absorption normalization, and typical-state Page turnover are explicit candidate results.
- Compute the microstate energy splittings and linewidths to decide whether the sector-centroid spacing produces resolvable Hawking lines or a dense continuum-like spectrum; retain the finite- n nonthermal entropy correction as the more robust target.

The candidate is falsified or requires major revision if no single microscopic update law can satisfy these conditions simultaneously, if refinement fails to recover the ADM/DeWitt universality class, if the finite-capacity strong-gravity branch is dynamically inconsistent, or if matter coupling requires incompatible causal/update structures.

41. Integrated conclusion

The construction developed in this paper can be summarized in four levels. First, the microscopic substrate is finite: bounded energy capacity and finite information are distinct, and the preferred record algebra is finite Weyl/symplectic. Exact consistency is assigned to finite update composition. A strict one-neighborhood-per- K BCC $d=4$ QCA provides a concrete linear propagation carrier with two positive-frequency amplitudes; the primitive law is the QCA itself, while a Hamiltonian is an infrared effective description. Second, smooth graph sectors recover geometry and the ADM/DeWitt deformation algebra. The inverse-metric H-H structure is generated by the state-dependent normal, while H-D and D-D recover their continuum forms under refinement. Third, high gravitational load is bounded. The Planck-density cap gives a core scale proportional to $M^{1/3}$, and the quadratic Compute-Link/chord-load completion produces a Hayward-type nonsingular profile. An explicit reversible Compute-Link circuit supplies a bounded correlation correction without altering the leading r^{-4} exterior power. Fourth, the horizon information sector is separated from the core: the rank-2 normal-current algebra gives a two-state effective boundary record, micro-IR matching assigns $\Delta A=4 \ln(2) l_P^2$, and a discrete horizon tower supports an exact energy-conserving unitary information-transfer map.

$$A_{micro} = 0, \quad A_{eff} \rightarrow 0 \quad \text{in the smooth infrared sector}$$

$$\rho \leq \rho_p, \quad r_c \propto M^{1/3}, \quad m(r) = \frac{Mr^3}{r^3 + \frac{3M}{4\pi\rho_p}} \quad \text{under the quadratic completion.}$$

The framework is not presented as a finished unique quantum theory of gravity. The remaining gaps are now more localized: the strict-local propagation QCA must be extended to the full nonlinear geometry/matter Hilbert space; Boundary Record Minimality must be derived or independently justified; and the absorption/greybody spectrum, exclusive microstate amplitudes, level splittings and linewidths must be calculated from that same update law. An independent phenomenological horizon-source constant is not required at the inclusive level, because microcanonical reciprocity relates emission to absorption through the horizon-state degeneracy ratio. Subject to these conditions, the paper now provides more than a static regular-core model: it contains a candidate microscopic boundary algebra, a sector-centroid horizon spectrum, a Hawking detailed-balance and normalization mechanism, causal evaporation scalings, and a typical-state Page turnover. These are concrete falsification targets rather than placeholders for future information-loss dynamics.

APPENDIX A. General p-channel derivation

Starting from $\rho/\rho_P = \chi^p$ with $\chi = 3m/(4\pi\rho_P r^3)$, the mass equation is

$$dm/dr = 3^p (4\pi\rho_P)^{1-p} m^p r^{2-3p}. \quad (A.1)$$

For $p \neq 1$, separation and integration give

$$m^{1-p} = C + \left[3/(4\pi\rho_P)\right]^{p-1} r^{-3(p-1)}. \quad (A.2)$$

For $p > 1$, $m(\infty) = M$ implies $C = M^{1-p}$. Factoring out M^{1-p} and defining $a^3 = 3M/(4\pi\rho_P)$ yields Eq. (9.2). The $p = 1$ case must be solved separately and gives m proportional to r^3 .

APPENDIX B. Positive-harmonic Weyl selection

Let $V(\delta) = \sum_m a_m [1 - \cos(m\delta)]$ with $a_m \geq 0$ and define $\chi_W = |\sin(\delta/2)|$. Saturation at $\delta = \pi$ gives $V(\pi) = 2 \sum_{m \text{ odd}} a_m$. Near the vacuum, δ approximately $2\chi_W$ and

$$V(\delta) = 2\chi_W^2 \sum_m m^2 a_m + O(\chi_W^4).$$

After normalizing by $V(\pi)$, exact infrared coefficient one requires Eq. (5.9). Since the difference between the two sides of Eq. (5.9) is

$$\sum_{m \text{ even}} m^2 a_m + \sum_{m \text{ odd } m>1} (m^2 - 1) a_m.$$

and every term is nonnegative, equality forces all coefficients except a_1 to vanish. The result is conditional on the nonnegative-harmonic assumption and the stated saturation/infrared normalization.

APPENDIX C. Numerical audit definitions

A.1 Weyl audit

For the finite-Weyl audit, X is the cyclic shift matrix and $Z = \text{diag}(1, \omega, \dots, \omega^{d-1})$. The reported residual uses $W_N = X^1 Z^2$ and $W_M = X^2 Z^1$, with the phase exponent $k = e - a - b - c$ evaluated modulo d .

A.2 One-dimensional finite-difference audit

The periodic interval has length 2π and lattice spacing $a = 2\pi/N$. The centered derivative is $(D_a f)_i = (f(i+1) - f(i-1))/(2a)$. The smooth test fields are

$$N(x) = 0.8 + 0.25\sin x + 0.10\cos(2x)$$

$$v(x) = 0.6 + 0.20\cos x - 0.08\sin(2x)$$

$$w(x) = -0.3 + 0.22\sin x + 0.12\cos(3x)$$

With $M_N = \text{diag}(N)$ and $L_v = \text{diag}(v)$ D_a , the residuals are

$$R_{HD} = [L_v, M_N] - \text{diag}(v D_a N)$$

$$R_{DD} = [L_v, L_w] - \text{diag}(vD_a w - wD_a v)D_a$$

Relative Frobenius norms are reported. Smooth-sector norms restrict the input to Fourier modes $|k| \leq 3$ while holding that physical mode range fixed as N increases.

A.3 Three-dimensional normal-refresh audit

The graph audit uses centered derivatives on the parameter cube and reconstructs a unit normal in R^4 from the three tangent records. Two outer layers are removed from norm estimates. After each normal update the tangents and normal are recomputed from the updated embedding. The fitted coefficient is the least-squares projection of $X_{NM} - X_{MN}$ onto $\epsilon_a h^{(ab)}(N \partial_b M - M \partial_b N)$.

APPENDIX D. Logical reading of exactness

Three notions of exactness should not be conflated. (i) Matrix exactness: finite Weyl and finite gauge operators obey exact matrix identities. (ii) Microscopic path consistency: finite scheduler compositions differ only by admitted finite gauge/address arrows. (iii) Continuum constraint exactness: the ADM/DeWitt HDA holds as an infinitesimal algebra. This framework assigns (i)-(ii) to the finite microscopic theory and requires (iii) only in the appropriate emergent smooth sector.

APPENDIX E. Compact architecture audit

Layer	Requirement	Current status
Microscopic kinematics	Finite local Hilbert spaces and bounded observables	Specified
Microscopic algebra	Finite Weyl and symplectic/Clifford carrier	Specified at candidate level
Microscopic closure	Exact finite update/groupoid composition	Established as design principle; exact in finite Weyl sector
Emergent geometry	Metric/triad records from finite relational data	Proof-of-principle
Infrared gravity	ADM/DeWitt HDA in smooth sector	Analytic target + numerical pilot support
Ultraviolet strong gravity	Finite capacity and Planck-density saturation	Derived at effective level
Regular black hole	Quadratic load law and Hayward-type profile	Derived conditionally
Quantum information	Rank-2 horizon algebra and unitary information transfer	Conditional candidate
Full dynamics	Unique 3+1 many-body Lorentzian update	Open
Phenomenology	Sector-centroid gaps, microstate splitting, and finite-n corrections	Conditional spectral prediction

APPENDIX F. Horizon counting and area-normalization audit

F.1 Finite geometry record and area spectrum

The horizon information count does not require each raw graph face to carry an integer multiple of l_P^2 . If a finite inverse-triad record E defines $G = E E^T$, then for a coordinate face with normal covector n the physical area is schematically

$$A_n(E) = l_P^2 \sqrt{(n_a G^{(ab)} n_b) / \det G}.$$

A finite geometry sector therefore gives a discrete area spectrum, but finite-Weyl kinematics alone does not fix the dimensionless spacing of that spectrum. Different admissible finite record maps can preserve the same infrared metric architecture while changing microscopic area eigenvalues.

F.2 Transfer-matrix formulation

For gauge-reduced boundary types α, β let $C_{(\alpha\beta)}$ be the number of admissible gauge-inequivalent boundary extensions and $a_{(\alpha\beta)}$ the corresponding dimensionless area increment. Define

$$(T_x)_{(\alpha\beta)} = C_{(\alpha\beta)} \exp[-x a_{(\alpha\beta)}].$$

At large area the state count grows with exponent x_* fixed by the Perron condition $\rho(T(x_*)) = 1$. The Bekenstein-Hawking density corresponds to $x_* = 1/4$, equivalently $\rho(T(1/4)) = 1$.

F.3 Uniform-area finite-state no-go

If every allowed step has the same nonzero algebraic area a_0 and C is a finite nonnegative integer matrix, then $x_* = \ln \rho(C)/a_0$. Exact $x_* = 1/4$ would require $\rho(C) = \exp(a_0/4)$. The Perron eigenvalue $\rho(C)$ is algebraic, whereas $\exp(a_0/4)$ is transcendental for nonzero algebraic a_0 . Hence a finite integer transition grammar with a uniform algebraic microscopic area cannot by itself produce the exact coefficient $1/4$. This no-go is one reason the present paper does not claim to derive the entropy coefficient from finite-Weyl state counting alone.

The route adopted in Part IV avoids this obstruction by separating information from geometry. The rank-2 boundary algebra fixes $\Delta S = k_B \ln 2$, while semiclassical horizon thermodynamics fixes the physical conversion $\Delta A = 4 \ln 2 l_P^2$. Nonuniform metric-weighted area spectra, refinement limits, or renormalized effective boundary grammars provide alternative routes but are not required for the minimal branch developed here.

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